

**K Controls Ltd**

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Process Valve Networking Monitoring and Control**EC Declaration of Conformity****1. K Controls Ltd of 2 Crown Way, Crown Business Centre, Horton Road, West Drayton, Middlesex, UB7 8HZ. England:**

Declare conformity with IEC 61508 – Functional Safety of Electrical/Electronic/ Programmable Electronic Safety Related Systems

2. We hereby declare that the 007 Series of Valve Position Monitor Switchboxes, Type M6, are suitable for both Low and High Demand scenario applications at SIL 1 and SIL 2 of IEC 61508:2010 in the following respects:**2.1 Random Hardware Failures and Safe Failure Fraction****Failing to provide the annunciated function**

		SFF %	LOW DEMAND			HIGH DEMAND	
			F rate/hr	PFD*	SIL	F rate pa	SIL
M6 - Ex d Microswitch	Closed (Fail to open)	74	0.018 10 ⁻⁶	7.9 10 ⁻⁵	2	1.6 10 ⁻⁴	2
M6 - Ex d Microswitch	Open (Fail to close)	24	0.052 10 ⁻⁶	2.3 10 ⁻⁴	1	4.6 10 ⁻⁴	1

* assumes an annual proof test interval

Spurious annunciation (no valve movement)

		F rate pa
M6	Closed (Spuriously opens)	4.5 10 ⁻⁴
M6	Open (Spuriously closes)	1.5 10 ⁻⁴

2.2 Life cycle documentation (Safety Plan, Safety Spec, Safety Manual, Validation report)**2.3 Appropriate control over purchased materials.****2.4 Independence of the review and assessment Technis report T549_5_1 (23.09.13)****2.5 The following conditions indicate the limits of application for the 007 – M6 Series:**

Temperature rating: -50°C min. to +55°C max (unless stated in applicable certification).

Switch safe operating parameters: - Do not exceed values shown on product label.

Maintenance Schedule: It is recommended that a periodic visual and operation evaluation is carried out at least once a year or every 1 million cycles.

Product life: 5 Million cycles.

3. Control of internal production compliance:

- BS EN ISO9001:2008 (National Quality Assurance Certificate 3723)

- ATEX Quality Notification No. Baseefa ATEX 0600 / IECEx QAS GB/BAS/QAR06.0081/04

4. Authorised Person for the Manufacturer within the European Community:

Name: Brian E Prince

Signature:

Position: Joint Managing Director

Date: 28.11.14